

Measure DC Voltage in a Circuit

Two-resistor voltage divider measured with the ADALM2000 (M2K) and Scopy Voltmeter

Circuit and measurement point

The circuit used a $510\ \Omega$ upper resistor and a $1.1\ \text{k}\Omega$ lower resistor. W1 drove the top of the divider, GND was at the bottom, and Channel 1 (1+) measured V_{OUT} at the resistor junction relative to 1-/GND.

Brief procedure

1. Configure W1 for the requested DC input. 2. Connect 1+ to V_{OUT} and 1- to GND. 3. Select DC mode in Scopy Voltmeter and record the displayed output voltage.

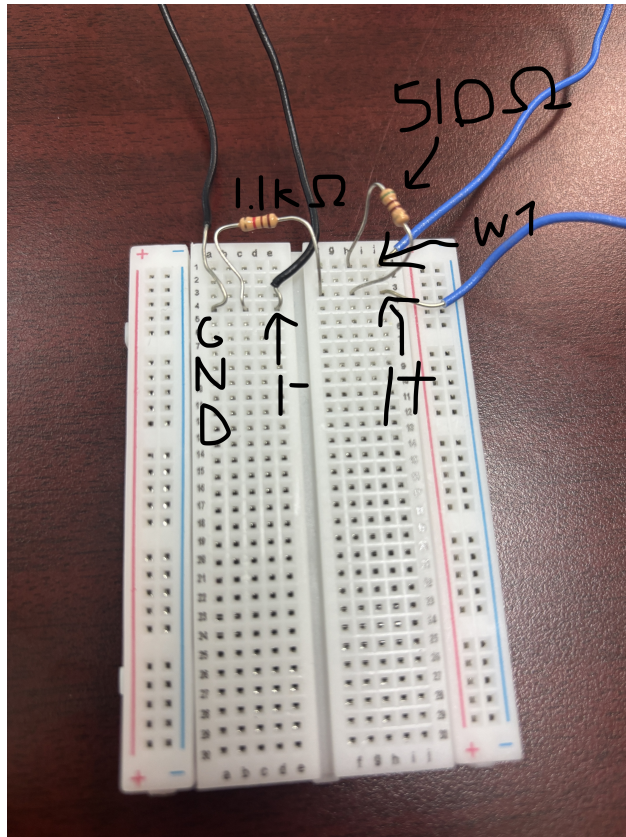


Figure 1. Labeled breadboard photo of the divider and measurement wiring.

Recorded DC results

W1 DC input	Scopy Voltmeter reading at VOUT
+5 V	+3.251 VDC
+2.5 V	+1.660 VDC
-5 V	-3.367 VDC

Scopy Voltmeter evidence



Figure 2. +5 V input: $V_{OUT} = +3.251$ VDC.

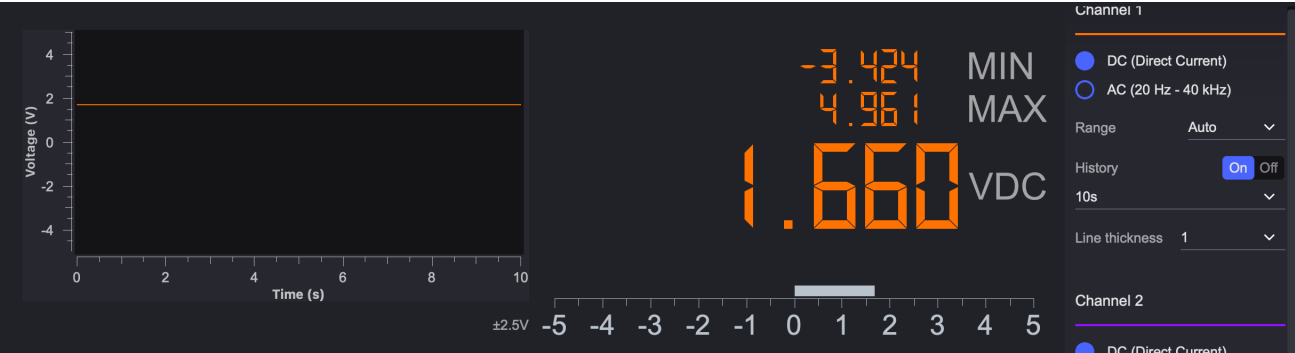


Figure 3. +2.5 V input: $V_{OUT} = +1.660$ VDC.

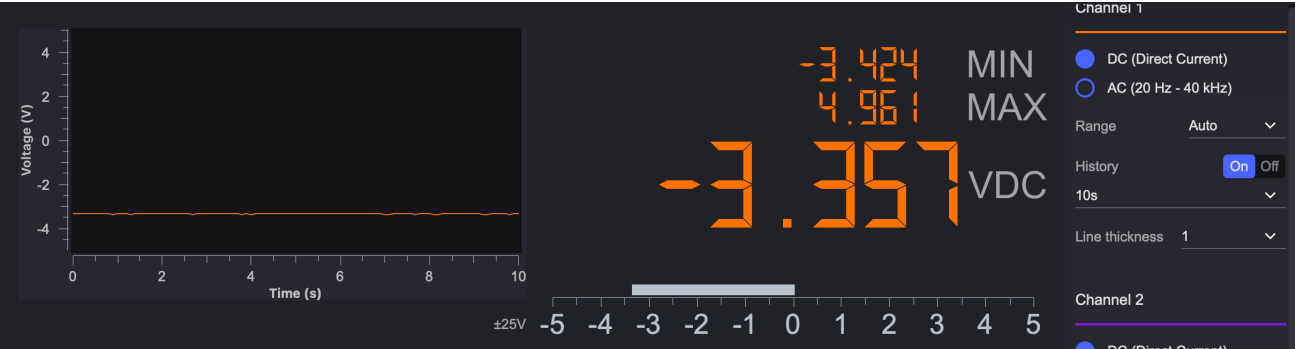


Figure 4. -5 V input: $V_{OUT} = -3.367$ VDC.